A.K. TECHNICAL UNIVERSITY, LUCKNOW

Syllabus 3rd Year
[Effective from session 2018-19]

B. Tech. in Plastic Engineering

A. K. TECHNICAL UNIVERSITY, LUCKNOW

STUDY & EVALUATION SCHNME
B. Tech. Plastic Engineering
Year 3rd, SEMESTER-V
[Effective From session 2018-19]

S. No.	Course code	Subject	Periods			Evaluation Scheme			ESE	Subject Total	Credit
			L	T	P	Sessional Assessment					
						CT	TA	Total			
Theory Subjects											
1.	RAS501	Engineering Economics	3	0	0	20	10	30	70	100	3
2.	RAS502/ RUC501	Sociology/ Cyber Security	3	0	0	20	10	30	70	100	3
3.	RPL501	Plastic Materials-1	3	0	0	20	10	30	70	100	3
4.	RPL502	Plastic Processing-1	3	1	0	20	10	30	70	100	4
5.	RPL503	Rheology & Testing of polymers	3	0	0	20	10	30	70	100	3
6.	RPL 051-053 RME502	Department Electives	3	1	0	20	10	30	70	100	4
Practical/Design/Drawing											
7.	RPL551	Analysis & Identification of Polymers Lab	0	0	2			50	50	100	1
8.	RPL552	Plastic Processing Lab	0	0	2			50	50	100	1
9.	RPL553	Synthesis & Polymerization Resins Lab	0	0	2			50	50	100	1
10.	RPL555	SEMINAR	0	0	2	-	-	100	-	100	1
		Total	**18**	**2**	**8**	120	60	430	570	**1000**	**24**

Departmental Elective I

RME502: Heat and Mass Transfer
RPL051: Chemical Thermodynamics
RPL052: Polymer Degradation and Stabilization
RPL053: Plastic Packaging & Foam

A. K. TECHNICAL UNIVERSITY, LUCKNOW

STUDY &EVALUATION SCHNME
B. TECH. Plastic Engineering
Year 3rd, SEMESTER-VI
[Effective From session 2018-19]

S. No.	Course code	Subject	Periods			Evaluation Scheme			ESE	Subject Total	Credit
			L	T	P	Sessional Assessment					
						CT	TA	Total			
Theory Subjects											
1.	RAS601	Industrial Management	3	0	0	20	10	30	70	100	3
2.	RUC601/ RAS602	Cyber Security/ Socialogy	3	0	0	20	10	30	70	100	3
3.	RPL601	Plastic Product and Mould Design	3	0	0	20	10	30	70	100	3
4.	RPL602	Plastic Materials-II	3	1	0	20	10	30	70	100	4
5.	RPL603	Plastics Processing-II	3	0	0	20	10	30	70	100	3
6.	RPL061-064	Department Elective Course-II	3	1	0	20	10	30	70	100	4
Practical/Design/Drawing											
7.	RPL651	Plastic Material Testing Lab	0	0	2			50	50	100	1
8.	RPL652	Plastic Product Testing Lab	0	0	2			50	50	100	1
9.	RPL653	Polymer Characterization	0	0	2			50	50	100	1
10.	RPL654	Computer aided Product & mould Design Lab	-	-	2			50	50	100	1
		Total	**18**	**2**	**8**	120	60	380	620	**1000**	**24**
		Industrial Training-II of 4-6 weeks after VI Sem will be evaluated in VII Sem									

Departmental Elective II

RPL061: Computational Fluid Dynamics
RPL062: Chemical Reaction Engineering
RPL063: Additives and Compounding
RPL064: Fiber Technology

RPL-501: PLASTIC MATERIALS – I **L: T: P**
3: 0: 0

Unit – I **8**

Introduction to Polymer Chemistry

Sources of raw materials- Monomers - Polymers- Polymerization - Types of Polymerization – Classification.Definition and Classification of Plastics – General properties – Historical development of plastic industry-future trends, Thermoplastics, Thermosetting, Engineering & High performance plastics.

Unit –II **8**

Structure of Plastics: Molecules –Crystallinity – Effect of Crystallinity on properties – crosslinked plastics – Determination of Molecular weight – Effect of Molecular weight on processing and properties – Molecular weight distribution. Linear, Branched and cross linked structures in polymers. Flexibility and movement of macromolecules. Glass transition temperature (Tg). Relationship

Unit –III **8**

Introduction- Sources and manufacture of raw materials- basic chemistry- Methods of manufacture- Flow behavior- General properties and applications of Olefin Polymer and Co- polymers Vinyl chloride polymers and co-polymers

Unit –IV **8**

Introduction – Sources and manufacture of raw materials –basic chemistry – Methods of manufacture –Flow behavior – General properties and applications of Styrene and Styrene co- polymers PMMA. Cellulose polymers

Unit –V **8**

Bio degradable and Bio Plastics

Principle and Mechanism of Plastics degradation, Natural Bio-degradable Polymers – Synthetic Bio-degradable Polymers – Water soluble Polymers. Bio plastics types, properties and applications

Case studies on applications of above materials.

Reference Books –

1. Brydson, J.A “Plastics Material”
2. Schwartz & good man “Plastics materials and processing”
3. Irwin rubin “Hand book of Plastic Materials and Technology”

RPL-502: PLASTIC PROCESSING-I **L : T: P**
3 : 1: 0

Unit – 1 **06**

Processability of various polymers and the role of rheology in polymer processing.

Unit – 2 **09**

General description of extrusion processes, type of extruders, screw and their output in terms of drag, leakage and pressure flow, influence of screw dimensions and output, die and screw characteristics. Design of barrel and screw for commodity, heat sensitive and engineering polymers.Barrier Screws.

Unit – 3 **09**

Individual extrusion systems, Dies, Sizing and Downstream equipments, Faults, Causes and Remedies for film, pipe, lamination, profiles, cables, sheet, Box Strapping .

Unit – 4 **08**

Twin-screw extrusion and Co Extrusion systems.Casting of films.Multi layer systems for Films and

Pipe . Faults , Causes & Remedies.

Unit – 5 **08**

General description of Compression and Transfer moulding and its application in processing of thermosetting materials.Faults , Causes & Remedies.

Books –

1. Berins , “Plastics engineering hand book” - society of the Plastics industry
2. Allen , WS & baker , pn “Hand book of Plastics technology
3. Chris Rauwendaal, “Polymer Extrusion” Hanser Publication
4. Iyesew, A.I “Compression molding”
5. BOBB “COMPRESSION & TRANSFER THEORY & TNCHNOLOGY”

RPL-503: RHEOLOGY AND TESTING OF POLYMERS **L : T: P**
3 : 0: 0

Unit – 1 **8**

Introduction to polymer rheology, Newtonian and non-Newtonian fluids.Viscoelastic behaviour, dynamic and constitutive equations.Mechanical models.Discussion of models for flow and deformation in polymers and treatment of measureable rheological properties.

Unit – 2 **7**

Measurement of viscosity and normal stresses.Viscous heat generation Interpretation of time-temperature senstivity of viscoelastic solids and liquids.

Unit – 3 **9**

Chemical Properties - Cohesive energy, cohesive energy density, solubility parameter, Prediction of solubility parameter, Effect of polymer structure on solubility in solvents and oils, Influence of structure in prediction of flame retardancy, water repellency , Chemical resistance of polymers, Polymer-in-solution: Polymer-solvent interaction, good and poor solvents, intrinsic viscosity and Mark-Houwink equation. Concept of fractionation processes.

Unit – 4 **8**

ASTM, BS, IS and its importance, correlation of these tests with actual performance. Degradation behavior of polymers.Introduction to polymer characterization by advance instrumental techniques such as IR, NMR, DSC, TGA, SEM, etc.

Books

1. Vishu shah “Plastics testing technology hand book”
2. Brown RP “Hand book Plastics test methods “
3. Mathur ,AB &Bhardwaj IS “Testing and evaluation of Plastics”
4. Ives GC, Mead JA & Riley MM “Hand book Plastics test methods”
5. Braun “simple methods for identification of Plastics”
6. Crompton “Analysis of Plastics”
7. Plastics engineering hand Ed 5 Society of the Plastics industry
8. Haslam“identification and analysis of plastics”
9. Allen , WS & Baker , PN “Hand book of plastics technology 2 vols”

RPL-551 ANALYSIS AND IDENTIFICATION OFPOLYMERS LAB **L: T: P**
0: 0: 2

Minimum eight experiments out of the following:

1. Identification of unknown polymer using heating, burning, solubility.
2. Confirmatory chemical tests for Identification of unknown polymer .
3. Quantitative estimation of the basic raw materials and auxiliaries used in polymer such as phenol, urea, formaldehyde.
4. Quantitative estimation of the basic raw materials and auxiliaries used in polymer such as glycerol, plasticizer's initiators.
5. Quantitative estimation of the basic raw materials and auxiliaries used in polymer industries such as antioxidants, etc
6. Determination of purity of solvents, monomers and other auxiliaries.
7. Determination of physical properties - boiling point using standards techniques.
8. Determination of physical properties - melting point.
9. Determination of physical properties - refractive index.
10. Determination of physical properties - specific gravity of polymer materials .

RPL552: PLASTIC PROCESSING LAB **L : T: P**
0 : 0 : 3

Minimum eight experiments out of the following:

1. Auto Injection Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
2. Micro-Processor Controlled Injection Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
3. Extrusion Process – Free sketch of Machine, Study of Parts & their function. Practice on Die setting, Cycle time analysis, Start up and shut down Procedure.

4. Compression Moulding or Transfer Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
5. Blow moulding Process – Free sketch of Machine, Study of Parts & their function, Parison die. Practice on Die centering, Cycle time analysis, Start up and shut down Procedure.
6. Thermoforming (Vacuum forming) Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
7. Rotational Moulding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
8. Plastics coating Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
9. Plastics Sealing Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
10. Plastics welding Process – Free sketch of Machine, Study of Parts & their function. Cycle time analysis, Start up and shut down Procedure.
11. Screen-Printing on Plastics
12. Hand lay Process for FRP – Study of resin and other components. Making of a product.

RPL553: SYNTHESIS & POLYMERIZATION RESINS LAB **L : T: P**
0 : 0 : 2

Minimum eight experiments out of the following:

1. Synthesis of a polymer by Bulk polymerization Techniques
2. Synthesis of a polymer by solution polymerization Techniques
3. Synthesis of a polymer by suspension polymerization Techniques
4. Synthesis of a polymer by emulsion polymerization Techniques
5. Determination of molecular weight by viscosity.
6. Preparation of phenol formaldehyde resin
7. Preparation of urea. formaldehyde resin
8. Preparation of unsaturated polyester resin
9. Determination of acid value in unsaturated polyester resin
10. Preparation of saturated polyester resin
11. Determination of acid value in saturated polyester resin
12. Synthesis of a copolymer based on any common monomers like styrene, acrylates, maleic anhydride, acrylic acid and methacrylic acid
13. Modification of epoxy resin modification of any natural polymer such as cellulose, rosin, natural rubber, etc. .
14. Depolymerization of waste thermoplastics such as polystyrene or polymethylmethacrylate .

RPL-601: PLASTIC PRODUCT & MOULD DESIGN L : T: P 3 : 1: 0

Unit – 1 **8**

Design of polymeric product. Design criteria based upon product functions and geometry. Material selection by property assessment.Selection of appropriate forming processes. Moulding considerations: Draft, radii, dimensional tolerances, wall thicknesses, ribs and bosses, inserts, sink marks, undercuts, feeding system, gate location, flow pattern, shrinkage and post moulding shrinkage.

Unit – 2 **8**

Design of Plastic under static load; Design of Plastic under Dynamic load - Gear, Bearing. Metal insert, hinge, fastners.

Unit – 3 **8**

Injection mould design: Single, multicavity, semi automatic and automatic moulds. Types of injection mould, their application, detailed structure and working. Feed system, Temperature control system, Ejection System, Standard Mould base.

Unit – 4 **8**

Split Mould and types of mechanism, Unscrewing mechanism, Introduction to Hot runner mould.

Design concepts for compression moulds, transfer moulds and blow moulds.

Unit – 5 **8**

Extrusion Dies - Types of extrusion dies and design characteristics. Die Design for Pipe and Sheet.

Books –

1. David H Morton Jons John wellis “Polymer product design materials and processing”
2. Rao NS “Design data for plastics engineers”
3. Rao NS “design formula for plastics engineers”
4. Joshi MV “dies for plastics extrusion”
5. Millar ,Edward “plastics product design part A& B”
6. DYM “product design with plastics”
7. Beck “plastics product design”
8. Dubois “plastics product design engineering hand book”
9. Malloy , Robert, A “plastics part design for injection moulding”
10. Bebb,R.H., “Plastics Mould Design Vol.1 Compression and Transfer Moulds
11. Pye R.G.W., “Injection MOULD Design for Thermoplastics”
12. DYM, “Injection Moulds and Moulding”
13. Lee NC “Blow moulding design guide”

RPL-602 : PLASTIC MATERIALS – II **L: T: P 3: 0: 0**

Unit –I **8**

Introduction – Sources and manufacture of raw materials – basic chemistry – Methods of manufacture – Flow behaviour – General properties and applications of -Polyamides – Nylon 6, 66 etc. ; Acetal-Homopolymer& Co-polymer Saturated polymers – PETP & PVTP ; Polymers containing fluorine – PTFE, PVDF etc.; Polycarbonate ; Thermoplastics Polyester, Poly urethane

Unit –II **8**

Introduction – Sources and manufacture of raw materials – basic chemistry – Methods of

manufacture – Flow behaviour – General properties and applications of Polyphenylenesulphide ;Polyphenylene oxide ; Polysulphones ; Poly ether ether ketone

Unit III **8**

Introduction – Sources and manufacture of raw materials – basic chemistry – Methods of manufacture – Flow behaviour – General properties and applications of Speciality Plastics, Plastic Composites, , High performance Polymer

Unit –IV **8**

Thermosetting plastics.

Introduction – Sources for raw materials – resin preparation – polymer structure additives – curing and cross linking–agents, processing behavior, general properties and applications of Phenol formaldehyde, Urea formaldehyde, Melamine formaldehyde, Unsaturated polyesters. Epoxide resins, Polyurethanes, Silicones, etc.

Unit – V **8**

Alloys and blends

Introduction to polymer blends & alloys- Definitions and nomenclature reasons for making polymer blend, compatibilizationmechanisms, Miscibility and flow behaviour of polymer blends, Immiscible blends,

Books –

1. Plastics Material by Brydson, J.A.
2. lrwinrubin “Hand book of Plastic Materials and technology”
3. Athalye ,AS “Plastics Materials hand book”
4. Gowariker VR “Polymer science”
1. Berins “Plastics engineering hand book” Society of the plastics industry Inc . SPI
2. Schwartz & Good man “Plastics materials and processing”
3. Biriey&scott “Plastics materials (properties and application)”
4. Billmeyer, FW, “Text book of polymer science” 5

Harper, “Modern of Plastics hand book”

RPL-603: PLASTIC PROCESSING-II **L : T: P : 3 : 0: 0**

Unit – 1 **8**

Basic concepts of injection moulding for thermoplastics.Machine layout, construction and specification, type of injection units. Principle and theory of standard operation, elements of moulding cycle, screw plasticizing and conveying output, screw driver principles, outline of mould features, clamping devices-hydraulic and toggle types.

Unit – 2 **8**

Process variables and their importance, temperature, pressure, injection rate, etc.Faults and remedies in injection moulding.Injection moulding of thermosets.Reaction injection moulding.

Unit – 3 **8**

Description of various thermoforming processes-simple vacuum, drape, bubble and plug assisted formings. Thermoforming and process variables affecting the product quality. Machining of Plastics

Unit – 4 **8**

General description of blow moulding processes, type of blow moulding machines, parison control, types of Dies, process variables, problems and their remedies. Stretch blow moulding

Unit – 5 **8**

Rotational moulding- description and features of rotational moulding and its comparison with blow moulding. Welding / Joining of Plastics – Defination, Principle of Working ; FRP Processes – Hand lay, Spray, Autoclave, Filament winding, Pultrusion , Matched mold – Principal & working. Faults and remedies.

Books-

1. Rubin , Irvin “Injection molding theory and practice”
2. Berins “Plastics engineering hand book” By society of the Plastics industry
3. Strong A Brent “Plastics materials and processing”
4. Rosato DV “blow molding hand book”
5. Allen , WS & Baker , pn “Hand book of Plastics technology”
6. Bolur P C, “A guide to injection moulding of plastics”
7. Whelan A& craft JL “Development in injection moulding-1”
8. Schwartz &Gooodman “Plastics materials & processing”
9. Athalye, “Injection moulding”
10. James L throne “Thermoforming”
11. Crawford, RJ and Throne,JL “Basic principles of rotational molding”
12. Bruin “Basic principles of rotational molding”
13. Bruin “Basic principle of thermoforming”
14. Bryce “Plastics injection molding”
15. Bown,John “Injection Moulding of plastic Components”
16. Beadle “Plastics Forming”

RPL 651: PLASTIC MATERIAL TESTING LAB **L : T: P 0 : 0: 2**

Minimum eight experiments out of the following:

1. Determination of Ash Content in plastics materials.
2. Determination of Moisture Content in plastics materials.
3. Determination of Filler content in plastics materials.
4. Determination of Melt flow index of plastics materials.
5. Determination of Optical properties of plastics materials.
6. Determination of Thermal Properties of plastics materials.
7. Determination of Electrical Properties of plastics materials.
8. Study of Weathering properties of plastic materials.
9. Determination of Density of plastic materials.

10. Compounding or Blending using two roll mill
11. Specimen preparation using contour cutter .
12. Determination of Bulk density for powder materials.

RPL652: PLASTIC PRODUCT TESTING LAB **L : T: P 0 : 0: 2**

Minimum eight experiments out of the following:

1. Testing of HDPE Pipes
2. Testing of UPVC Pipes
3. Testing of Water Storage Tanks
4. Testing of Films/Sheets,
5. Testing of HDPE/PP Woven Sacks/Tapes,
6. Testing of Baby Feed Bottles,
7. Testing of Milk Packing Pouches
8. Testing of Meter box Cover .
9. Testing of UPVC Pipe Fitting
10. Testing of Irrigation Product-Lateral
11. Testing of Irrigation Product-Emitters
12. Testing of Irrigation Product-Quick Coupled Pipes
13. Testing of FRP Sheets
14. Testing of PVC Conduit .

RPL-653: POLYMER CHARACTERIZATION LAB **L : T: P 0 : 0: 2**

Minimum eight experiments out of the following:

1. Determination of Melt Flow Index of different Plastics Materials and Grades,
2. Determination of molecular weight by viscometry.
3. Determination of K-value of PVC.
4. Study of rheological properties of concentrated polymeric solution by Brook field viscometer or Rheoviscometer under variable shear rates.
5. Characterization by Weight loss of common polymers by Thermogravimetric, Analysis, (TGA).
6. Characterization of Filler Content /Ash Content of common polymers by Thermogravimetric, Analysis, (TGA).
7. Characterization of Thermal stability of common polymers by Thermogravimetric, Analysis, (TGA).
8. Characterization by Melting Range of common polymers by Differential Scanning Calorimetry (DSC).
9. Characterization by Tg of common polymers by Differential Scanning Calorimetry(
10. Study of UV stabilization of polymer samples by UV-visible spectrophotometer.
11. Study of UV stabilization of polymer samples by UV-visible spectrophotometer.

RPL654: COMPUTER AIDED PRODUCT & MOULD DESIGN LAB

L : T: P
0 : 0 : 2

Design Minimum 4 mould using cad

1. Single Impression Two Plate mould
2. Multi Impression two Plate Mould
3. Three Plate Mould (Multi Impression)
4. Hand Mould For External Undercuts
5. Split Mould - Without Delayed Action
6. Split Mould - With Delayed Action
7. Mould for Internal Undercuts
8. Insert Mould
9. Mould Design for Internally Threaded
10. Components (Automatic Unscrewing)
11. Hot Runner Mould

Design Minimum 2 Product using cad

2. Product for Agriculture Application.
3. Product for Automobile Application.
4. Product for Medical Application.
5. Product for Household Application.
6. Product for Electronic Application
7. Design of Plastic Gear
8. Design of Plastic Bearing

DEPARTMENTAL ELECTIVES

Departmental Elective I

RPL 051.Heat and Mass Transfer
RPL 052.Chemical Thermodynamics.
RPL 053. Polymer Degradation and Stabilization RPL 014.Plastic Packaging & Foam.

RPL-051 HEAT & MASS TRANSFER L T P 3 1 0

UNIT-1

Introduction to Heat Transfer:

Thermodynamics and Heat Transfer. Modes of Heat Transfer: Conduction, convection and radiation. Effect of temperature on thermal conductivity of materials; Introduction to combined heat transfer mechanism.

2

Conduction :

General differential heat conduction equation in the rectangular, cylindrical and spherical coordinate systems.Initial and boundary conditions.

3

Steady State one-dimensional Heat conduction :

Simple and Composite Systems in rectangular, cylindrical and spherical coordinates with and without energy generation; Concept of thermal resistance.Analogy between heat and electricity flow; Thermal contact resistance and over all heat transfer coefficient; Critical radius of insulation.

3

UNIT-2

Fins:

Heat transfer from extended surfaces, Fins of uniform cross-sectional area; Errors of measurement of temperature in thermometer wells.

3

Transient Conduction:

Transient heat conduction; Lumped capacitance method; Time constant; Unsteady state heat conduction in one dimension only, Heisler charts.

5

UNIT-3

Forced Convection:

Basic concepts; Hydrodynamic boundary layer; Thermal boundary layer; Approximate integral boundary layer analysis; Analogy between momentum and heat transfer in turbulent flow over a flat surface; Mixed boundary layer; Flow over a flat plate; Flow across a single cylinder and a sphere; Flow inside ducts;

Thermal entrance region, Empirical heat transfer relations; Relation between fluid friction and heat transfer; Liquid metal heat transfer.

5

Natural Convection :

Physical mechanism of natural convection; Buoyant force; Empirical heat transfer relations for natural convection over vertical planes and cylinders, horizontal plates and cylinders, and sphere, Combined free and forced convection.

3

UNIT-4

Thermal Radiation :

Basic radiation concepts; Radiation properties of surfaces; Black body radiation Planck's law, Wein's displacement law, Stefan Boltzmann law, Kirchoff's law; ; Gray body; Shape factor; Black-body radiation; Radiation exchange between diffuse non black bodies in an enclosure; Radiation shields; Radiation combined with conduction and convection; Absorption and emission in gaseous medium; Solar radiation; Green house effect.

8

UNIT-5

Heat Exchanger :

Types of heat exchangers; Fouling factors; Overall heat transfer coefficient; Logarithmic mean temperature difference (LMTD) method; Effectiveness-NTU method; Compact heat exchangers.

3

Condensation and Boiling:

Introduction to condensation phenomena; Heat transfer relations for laminar film condensation on vertical surfaces and on outside & inside of a horizontal tube; Effect of non-condensable gases; Dropwise condensation; Heat pipes; Boiling modes, pool boiling; Hysteresis in boiling curve; Forced convection boiling.

3

Introduction to Mass Transfer:

Introduction; Fick's law of diffusion; Steady state equimolar counter diffusion; Steady state diffusion though a stagnant gas film.

2

Books:

2. Fundamentals of Heat and Mass Transfer, by Incroperra& DeWitt, John Wiley and Sons
3. Heat and Mass Transfer by Cengel, McGraw-Hill
4. Heat Transfer by J.P. Holman, McGraw-Hill
5. Heat and Mass Transfer by Rudramoorthy and Mayilsamy, Pearson Education
6. Heat Transfer by Ghoshdastidar, Oxford University Press
7. A text book on Heat Transfer, by Sukhatme, University Press.
8. Heat Transfer by Venkateshan, Ane Books Pvt Ltd
9. Schaum's outline of Heat Transfer by Pitts & Sisson McGraw-Hill
10. Heat and Mass Transfer by R Yadav, Central Publishing House

RPL 052: CHEMICAL THERMODYNAMICS L T P 3 1 0

Unit-I **8**

Basic Concept
The first law and conservation of energy.Applications to steady, non steady flow and other engineering problems.The second law.Applications to engineering problems relating to equilibrium, maximum and minimum work.

Unit-II **8**
Properties of Pure Substances
Changes in thermodynamic properties and their inter-relationships.The ideal gas.Fugacity and Fugacity coefficients for real gases.

Unit-III **8**
Multicomponent System
Partial molal properties. Mathematical models for the chemical potential. Ideal and non-ideal solutions.Activity and activity coefficients.The Gibbs Duhem equations.Excess properties of mixtures.

Unit-IV **8**
Phase Equilibria
Criteria for equilibrium between different phases in Multicomponent no reacting systems.Applications to systems of engineering interest, particularly to vapour – liquid equilibria and solubility.

Unit-V **8**
Chemical Equilibrium
The equilibrium constant and the variation of yield in chemical reactions with pressure, temperature and Composition.

Reference Books
1. Y.V.C.Rao," Chemical Engineering Thermodynamics" University press (1997).
2. Smith, J.M., Van Ness, H.C. & Abbot, M.M. "Intro to Chemical Engineering
3. Thermodynamics ", 5th edition. New York: Mc-Graw Hill (1996)"
4. Daubert T.E., "Chemical Engineering Thermodynamics" McGraw Hill (1986).

RPL 053: POLYMER DEGRADATION AND STABILISATION L T P 3 1 0

UNIT I **8**

Introduction and Thermal Degradation: Definition - Modes of Polymer Degradation - Mechanistic Aspects - Single Step Process and Chain Reactions - Auto Oxidation - Random and Specific Site Attack - Thermal Degradation: Introduction - Methods for Evaluation of Heat Resistance (DTA, DSC, TGA, TMA) - Mechanistic Aspects - Heat Resistance Polymers - Ablation –Stabilization – Thermal Degradation and Recycling - Heat Effect in Bio Polymers.

UNIT II **8**
Mechanical Degradation and Ultrasonic Degradation: Introduction - Mechanistic Aspects - Degradation Studies - Polymer Degradation in Solution. Ultrasonic Degradation - Importance - Experimental Methods - Mechanism of Ultrasonic Degradation (Cavitation and Direct Effects) - Degradation Studies (Detection of Transient Species and Molecular Weight Distribution) Application of Mechanical Degradation: Stress - Induced Chemical Alterations of Polymers- Mastication of Natural and Synthetic Rubber - Mechano Chemical Synthesis of Block and Craft Copolymers.

UNIT III **8**

Photo degradation: Introduction - Mechanistic Aspects (Excited States, Free Radicals and Ionic Species, Energy Transfer and Energy Migration) - Degradation in the Absence of Oxygen (Norrish Types I & II Reactions) - Photo Oxidation (Auto Oxidative Process, Sensitized Degradation) - Stabilization - Application: Polymers with Predictable Life Time, Photo resists.

UNIT IV **8**

Degradation By High Energy Radiation and Biodegradation:Introduction - Aspects of Radiation - Mechanistic Aspects - Simultaneous Cross Linking and Degradation - Radiation Stability and Protection Radiation Effects in the Bio Polymers - Application: Lithography, X - ray Resists in Contact Microscopy- Graft and Block Copolymerization Bio degradation - Modes of Biological Degradation - Enzymatic Degradation in Bio Polymers (Polysaccharides, Proteins, Malice Acids)

- Microbial Degradation of Synthetic Polymers - General Applications of Bio Degradable Plastics - Examples of Biodegradable Polyesters and Polyamides.

UNIT V **8**

Chemical Degradation: Introduction - Solvolysis - Polymer Characterization by Solvolysis - Stability of Polymer Against Solvolytic Agents - Commercial Applications - Ozonisation - Oxidative Degradation - Auto Oxidation of Polymers. Ionic Degradation: Alkaline Degradation of Poly Saccharides Acidic Degradation of Polyaldehydes and Polyacetals and Cationic Degradation of Polypropylene Sulphide and Polyesters.

BOOKS

1. W. Schnabel, Polymer Degradation - Principles and Practical Applications Hanser Publishers, New York, 1992.
2. Ann - Christine Albertsson, Samuel J. Huang, "Degradative Polymers Recycling and Plastic Waste Management" Marcel Dekker, New York, 1995.

RPL054: Plastic Packaging and foam **L T P**

3 1 0

UNIT-1 **8**

Introduction to packaging.Scope and functions of a package, advantages and disadvantages of polymeric packages over conventional packing materials.

UNIT-2 **8**

Forms of packages. Selection criteria of suitable polymeric packages for oils, fats and allied products, soaps and detergents, cosmetics, food, dairy products, beverages, medicines, chemicals, paints, household and industrial goods, etc. printing on polymeric packages.

UNIT-3 **8**

Testing, quality control and development in polymeric packaging.

UNIT-4 **8**

Introduction to rigid foam. Chemistry & physical formation, forming ingredients, their effect on foam morphplogy and physical properties of cellular plastics.

UNIT-5 **8**

Polymethane foam (rigid & flexible) pol expanded polystyrene foams, epoxy foam. Recent advance of foams.

BOOKS

1. Susan E.M. Seleke, "Understanding Plastic Packaging Technology", Hanser Publications – Munich
2. A.S. Altalye, "Plastics In Packaging", Tata Mcgraw–Hill Publishing Co. Ltd., New Delhi.
3. Mechanics' Of Cellular Plastics By Hilyard
4. Plastics Materials By Brydson
5. Hand Book Of Polymeric Foams & Foam Technology By Klempner

Departmental Elective II

RPL 061: Computational Fluid Dynamics
RPL 062: Chemical reaction Engineering
RPL 063. Plastic Extrusion Technology.
RPL 064. Fiber Technology.

RPL 061 : COMPUTATIONAL FLUID DYNAMICS **L T P 3 1 0**

Unit I **6**

Basic Concepts of Fluid Flow: Philosophy of computational fluid dynamics, conservation principles of mass, energy, and momentum, simplified flow models such as incompressible, inviscid, potential and creeping flows, classification of flows.
Turbulence and its Modelling: Transition from laminar to turbulent flow, Effect of turbulence on time-averaged Navier-Stokes equations, Characteristics of simple turbulent flows, Free turbulent flows, Flat plate boundary layer and pipe flow, Turbulence models, Mixing length model, The k-e model, Reynolds stress equation models, Algebraic stress equation models

Unit II **7**
Grid Generation: Structured and unstructured grids, choice of grid, general transformation of equations, some modern developments in grid generation in solving the engineering problems.
Finite Difference Method: Discretization of ordinary and partial differential equations, approximation of first, second and mixed derivatives, implementation of boundary conditions, discretization errors, applications to the engineering problems.

Unit III **10**
Finite Volume Method: Discretization methods, approximations of surface integrals and volume integrals, interpolation and differentiation practices, implementation of boundary

conditions, applications to the engineering problems. Introduction, one-dimensional steady state diffusion, two-dimensional diffusion problems, three-dimensional diffusion problems. The Finite Volume Method for Unsteady Flows and Implementation of Boundary Conditions: One- dimensional unsteady heat conduction,

Unit IV **7**
Special Topics: Flow in a sudden pipe contraction / expansion, flow and heat transfer in a complex tubes and channels, reactive flow, multiphase flow , and turbulent flow processes.

Suggested Books:

- Anderson Jr J. D., “Computational Fluid Dynamics: The Basics with Applications”, McGraw Hill. 1995
- Muralidhar K. and Sundararajan T., “Computational Fluid Flow and Heat Transfer”, Narosa Publishing House. 2003
- H. K. Versteeg and W. Malalasekera, An introduction to computational fluid dynamics: the finite volume method , Longman scientific & technical publishers 2007
- Ranade V. V, “Computation Flow Modeling for Chemical Reactor Engineering”, Academic Press. 2002

RPL 062: Chemical Reaction Engineering **L T P**

3 1 0

Unit I **8**

Rate of Reaction, Elementary and non-elementary homogeneous reactions, Molecularity and order of reaction, Mechanism of reaction, temperature dependency from thermodynamics, collision and activated complex theories. Integral and differential methods for analyzing kinetic data, interpretation of constant volume reactor, zero, first, second and third order reactions, half life period, irreversible reaction in parallel and series, catalytic reaction, auto catalytic reaction, reversible reactions.

Unit II **8**

Interpretation of variable volume batch reactions for zero, first and second order reactions, design equation for batch, continuous stirred tank, plug flow reactors for isothermal reaction.

Unit III **8**

Optimum reactor size, plug flow/mixed flow reactors in series and parallel, recycle reactor

Unit IV 8

Design of reactors for multiple reactions, parallel and series reactions. Temperature and pressure effects for single reaction.

UNIT V **8**

Residence time distribution of fluids in vessels, E, F and C curves, Dispersion model, Tank in series model. Non Isothermal PFR and CSTR, Safety issues in Non Isothermal Reactors.

Books:

1. Smith, J, M, “Chemical Engineering Kinetics”, 3rd Edition, McGraw-Hill (1990).
2. Levenspiel, O., “Chemical Reaction Engineering”, 3rd Edition, John Wiley (1998)

Reference Book: 1. Keith J. Laidler, “Chemical Kinetics” 3rd Edition, Pearson (2013)

RPL063: ADDITIVES & COMPOUNDING LTP : 3 1 0

Unit –I **8**

Additives for plastics

Types & Mechanism of Additive: - Fillers, Antioxidants, Thermal Stabilizers, Lubricants, Plasticizers, Toughening-agents, Colourants –dyes & pigments, Fire retardants, Coupling agents, blowing agents, Ultraviolet stabilizer, Antistatic agents, Anti blocking agents, Slip and antislip agents, processing aids, mould releasing agents.

Unit- II **8**

Compounding & Mixing – Selection of polymers and compounding-ingredients-general objectives-possibilities and limitation of additives into polymer matrices. Distributive and Dispersive mixing.

Unit – III **8**

Mixing equipments.

Intensive and Extensive Mixers – Machine construction – specifications – temperature control system – operating characteristics. Pelletizers.

Unit – IV **8**

Case studies on preference of one plastics to other and co-relation of properties of conventional materials and blends and alloys – case studies on application of blends and alloys.

Unit – V **8**

End Use Market for Plastics

Principles of Material selection including consideration of conventional materials competitive with plastics – Case studies on material suitability (e.g. Plastics Gears, Feeding Bottle, Bowels for micro wave ovens). Survey and uses of plastics with reasons for their importance in major industries like, Agriculture, Packaging, Building, Transport, Electrical, electronics and Telecommunications, Medical and Furniture.

Reference Books

2. Al – Malaika; S. Golovoy; A and Wilkie(Eds), Chemistry and Technology of Polymer Additives, Black well Science Ltd, Oxford (1999)
3. Matthews; F. L. and Rawlings; R.D., Composite Materials, Engineering and Science Chairman and Hall, London (1994)
4. Plastics Testing Technology Hand Books by Vishu Shah
5. Hand Book of Plastics Test Methods by Brown R. P
6. Mascia; L., The Role of Additive in Plastics, Edward Arnold Publishers Ltd., U. K. (1974).
7. Murphy; John, Additives for Plastics Handbook, 2nd Edition, Elsevier Advanced Technology, Oxford
8. Polymer additive – by Muller

RPL 064: FIBRE TECHNOLOGY **L T P 3 1 0**

UNIT-I

Introduction to natural and synthetic polymers.Essential characteristics and molecular architecture of fibre forming polymers

UNIT-II

Concept of order in polymers, crystallinity, orientation, physical structure of natural and man- made fibers.Physical methods for investigating fiber structure.Optical properties of oriented polymers and fibres, refractive index and birefringence.

UNIT-III 8

Melt spinning, dry and wet spinning of fibers. Fiber drawing, heat setting, texturing and mechanical properties of fibers based on viscose, cellulose acetate, polyamides.

UNIT-IV **8**

Fiber drawing, heat setting, texturing and mechanical properties of fibers based on polyesters, acrylics, polypropylene, glass and carbon-fibres. General principles of finishing and dying of fibers. Common types of finishes applied to textile fibers.

BOOK

A.A. Vaidya, Production of synthetic fibres, Prentice Hall of India Pvt. Ltd., New Delhi, 1988..